

TETOUAN, Morocco

WMO: 603180

Lat: 35.594N

Lon: 5.320W

Elev: 10

StdP: 14.69

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 42.7	(c) 45.0	(d) 33.6	(e) 28.2	(f) 55.7	(g) 35.9	(h) 30.9	(i) 54.0	(j) 30.0	(k) 59.2	(l) 27.7	(m) 59.6	(n) 5.9	(o) 220	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.1	(c) 92.2	(d) 68.2	(e) 88.9	(f) 68.4	(g) 85.9	(h) 68.4	(i) 75.4	(j) 81.5	(k) 74.4	(l) 80.3	(m) 73.4	(n) 79.2	(o) 12.6	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	125.2	78.4	72.2	119.3	77.6	71.5	116.3	77.2	38.8	80.9	37.8	80.4	36.9	79.5	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 26.3	(b) 23.2	(c) 20.9	(e) 37.5	(f) 98.7	(g) 3.4	(h) 4.5	(i) 35.0	(j) 102.0	(k) 33.0	(l) 104.6	(m) 31.1	(n) 107.1	(o) 28.6	(p) 110.4		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			36.0	77.0	2.7	2.3	34.0	78.6	32.5	79.9	30.9	81.2	28.9	82.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.4	56.9	57.3	59.8	62.5	66.8	73.2	77.4	78.6	74.4	68.9	62.4	58.2
	DBStd	8.49	3.91	3.82	3.64	3.31	3.81	3.36	3.51	3.22	3.52	3.63	4.42	3.92
	HDD50	4	1	2	0	0	0	0	0	0	0	0	0	1
	HDD65	1064	251	217	165	88	23	0	0	0	0	9	100	211
	CDD50	5997	216	206	304	374	522	695	850	887	731	585	371	256
	CDD65	1584	1	0	4	13	79	246	385	422	282	129	22	1
	CDH74	10077	0	1	23	42	285	1329	3099	3459	1478	335	25	1
	CDH80	2885	0	0	2	3	44	309	1039	1132	312	43	1	0
(14) Wind	WSAvg	10.1	10.3	11.3	11.2	11.0	10.1	10.7	10.2	9.5	9.4	8.2	10.0	9.5
(15) Precipitation	PrecAvg	26.9	4.1	3.8	3.1	2.6	1.4	0.5	0.0	0.0	0.8	2.6	3.7	5.1
	PrecMax	51.1	17.9	14.1	12.6	7.9	4.9	4.1	0.3	0.3	2.6	8.2	10.8	17.9
	PrecMin	10.7	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.2	0.0	0.0
	PrecStd	9.9	3.9	3.4	2.5	2.0	1.3	0.8	0.1	0.1	0.8	2.3	3.0	3.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	71.6	77.2	78.6	84.8	91.5	97.2	96.9	91.6	85.6	77.2	71.4
		MCWB	57.9	58.6	60.0	61.5	65.3	68.4	69.1	67.2	67.1	65.6	62.2	59.0
	2%	DB	67.5	68.0	71.6	74.3	80.2	86.4	92.2	92.6	86.2	80.4	73.7	68.2
		MCWB	57.8	57.5	59.2	61.6	64.5	66.8	68.6	68.1	67.8	66.1	63.1	58.8
	5%	DB	65.3	65.8	68.7	71.5	77.0	83.8	88.5	88.8	83.2	77.3	71.5	66.3
		MCWB	57.0	56.6	58.7	60.2	63.9	66.6	68.4	68.8	68.1	66.5	62.4	58.6
	10%	DB	63.3	63.9	66.3	69.0	74.0	80.7	85.6	85.7	80.8	75.4	69.5	64.6
		MCWB	56.3	56.1	58.0	59.8	63.3	67.0	68.8	69.2	69.1	66.3	61.6	57.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.6	61.0	63.3	65.5	69.4	73.3	76.5	77.5	74.5	71.2	67.5	62.6
		MCDB	66.2	68.1	70.7	74.8	77.8	81.2	84.0	82.8	80.8	77.6	73.5	67.1
	2%	WB	59.6	59.3	61.5	63.2	67.4	71.9	74.9	76.0	73.2	69.6	65.3	61.0
		MCDB	64.3	64.4	68.0	70.3	74.9	79.1	82.0	81.5	78.7	74.8	71.0	65.7
	5%	WB	58.7	58.5	60.1	62.1	66.0	70.8	73.7	74.9	72.3	68.7	63.9	59.8
		MCDB	63.3	63.2	66.1	68.5	72.9	77.7	80.5	80.2	77.8	73.8	69.1	64.5
	10%	WB	57.4	57.5	59.1	60.9	65.0	69.7	72.7	74.0	71.4	67.5	62.8	58.7
		MCDB	62.4	62.3	64.5	66.8	71.3	76.3	79.2	79.4	76.8	72.6	67.8	63.5
(35) Mean Daily Temperature Range	MDBR		12.5	12.4	12.4	12.4	13.0	14.1	14.8	14.1	12.9	12.5	12.4	12.7
	5% DB	MCDBR	14.1	15.7	17.1	16.3	18.0	19.7	20.8	19.7	16.6	15.6	14.2	13.6
		MCWBR	8.5	9.3	9.8	8.8	8.6	9.4	9.3	8.9	7.3	7.3	7.7	8.2
	5% WB	MCDBR	11.4	12.6	13.7	13.7	13.8	13.8	13.6	11.4	11.0	11.7	10.9	12.1
		MCWBR	7.9	8.6	8.8	8.5	7.8	7.7	7.4	6.2	5.7	6.5	6.6	8.2
(40) Clear-Sky Solar Irradiance	taub		0.336	0.364	0.412	0.410	0.430	0.458	0.472	0.477	0.443	0.407	0.367	0.343
	taud		2.501	2.414	2.258	2.293	2.264	2.192	2.139	2.138	2.264	2.366	2.455	2.499
	Ebn at Noon		273	277	271	278	272	264	259	255	258	258	259	263
	Edh at Noon		26	32	41	41	43	46	49	48	40	33	28	25
(44) All-Sky Solar Radiation	RadAvg		840	1083	1425	1765	2075	2382	2403	2117	1660	1212	891	748
	RadStd		79	134	154	117	158	88	47	82	81	109	74	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+151	+142

Nomenclature: See separate page